Dataset Doctor Analysis Report

Dataset: Test Dataset

1. Dataset Overview

Item	Value
Number of Rows	5,000
Number of Columns	32
Missing Values	0
Duplicate Rows	0
Target Column	age_of_driver
Problem Type	Regression

2. Data Quality

Quality Measure	Result
Overall Quality Score	80 / 100
Missing Values	Excellent
Duplicate Rows	Excellent
Data Types	Excellent
Correlation	Poor
Outliers	Poor

3. Target Balance

The selected target classes are reasonably balanced.

Class	Proportion
-1	0.1484
30	0.0240
22	0.0222
27	0.0206
42	0.0202
36	0.0202
28	0.0198
31	0.0196
32	0.0194
24	0.0194
25	0.0194
23	0.0190
40	0.0184
39	0.0184

21	0.0184
38	0.0182
37	0.0178
29	0.0176
26	0.0176
57	0.0170
44	0.0168
41	0.0168
20	0.0164
34	0.0164
18	0.0164
35	0.0162
19	0.0162
54	0.0158
51	0.0156
33	0.0156
45	0.0150
47	0.0142
55	0.0138
48	0.0138
60	0.0136
58	0.0136
43	0.0136
52	0.0126
49	0.0122
56	0.0118
59	0.0114
46	0.0114
53	0.0114
50	0.0112
61	0.0108
17	0.0102
62	0.0100
63	0.0088
64	0.0068
66	0.0066
65	0.0062
67	0.0054
70	0.0052
68	0.0048
74	0.0048

72	0.0044
69	0.0044
16	0.0044
71	0.0044
76	0.0040
75	0.0038
77	0.0038
73	0.0036
82	0.0030
78	0.0028
80	0.0024
15	0.0020
14	0.0020
79	0.0020
83	0.0020
81	0.0018
10	0.0014
11	0.0014
13	0.0012
87	0.0010
90	0.0010
85	0.0010
12	0.0008
84	0.0008
8	0.0006
89	0.0006
86	0.0006
93	0.0004
92	0.0004
88	0.0004
9	0.0002
4	0.0002
99	0.0002

4. Machine Learning Results

Model	MAE	RMSE	R2
Random Forest	1.9036	2.5445	0.9859
Linear Regression	4.5172	5.6111	0.9316
Decision Tree	2.4980	3.4753	0.9738

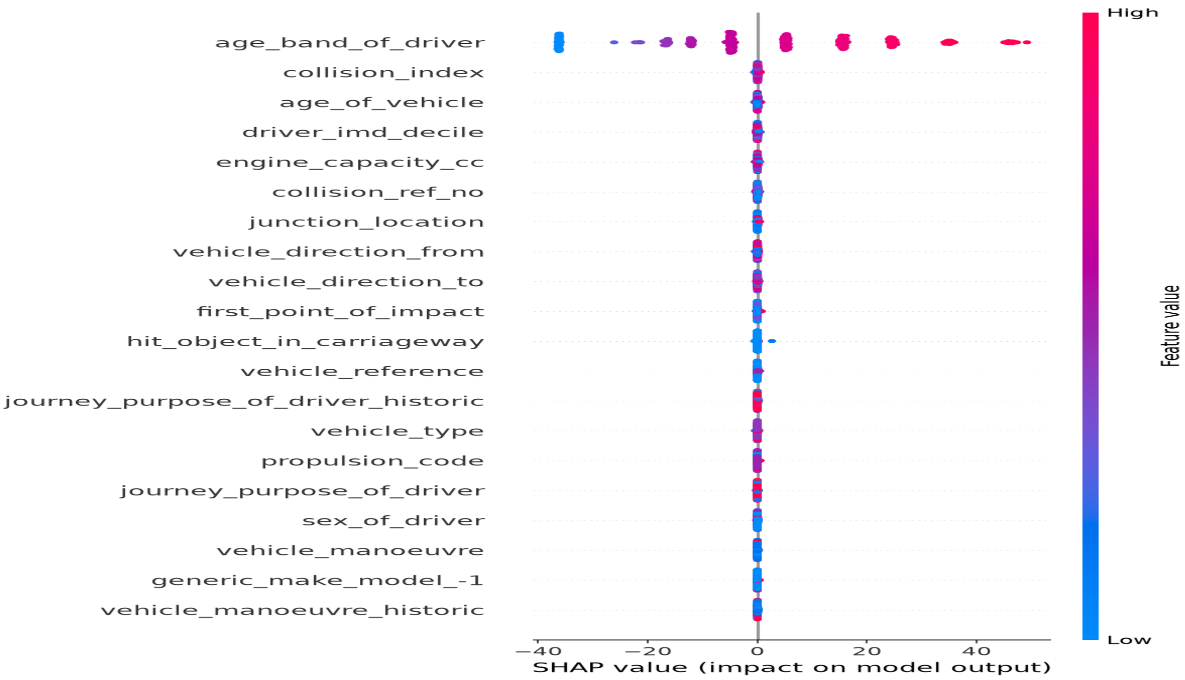
Best Model: Random Forest
R2: 0.9859

5. Explainable AI / SHAP

SHAP was used to explain which features had the strongest influence on the predictions made by the selected best-performing model.

Rank	Feature	Mean |SHAP Value|
1	age_band_of_driver	17.4254
2	collision_index	0.1042
3	age_of_vehicle	0.0843
4	driver_imd_decile	0.0771
5	engine_capacity_cc	0.0759
6	collision_ref_no	0.0724
7	junction_location	0.0515
8	vehicle_direction_from	0.0505
9	vehicle_direction_to	0.0495
10	first_point_of_impact	0.0454

SHAP Summary Visualization



6. Conclusion

The selected target column was 'age_of_driver'. The task was identified as a regression problem. The best-performing model was Random Forest, with a R2 score of 0.9859. The most influential features identified by SHAP were age_band_of_driver, collision_index, age_of_vehicle.